Geometric Sequences and Series

Section 1: General Term of a Geometric Sequence

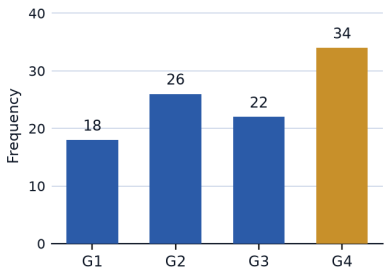
1 State the formula for the n-th term a_n of a geometric sequence with first term a_1 and common ratio r.

a. $a_n = a_1 r^{n-1}$

b. $a_n = a_1 r^n$

c. $a_n = a_1 + (n-1)r$

d. $a_n = \frac{a_1}{r^{n-1}}$



2 Find the 6th term of the geometric sequence $3, 6, 12, 24, \ldots$

a. 96

b. 192

c. 48

d. 128

3 In a geometric sequence, $a_2 = 12$ and $a_5 = 324$. Find the common ratio r.

a. 3

b. 2

c. 4

d. $\sqrt{3}$

Section 2: Finite Geometric Series

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 State the formula for the sum S_n of the first n terms of a geometric series ($r \neq 1$).

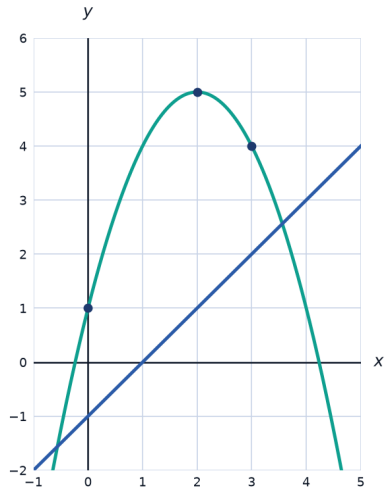
a. $S_n = \frac{a_1(1-r^n)}{1-r}$

b. $S_n = \frac{a_1}{1-r}$

c. $S_n = \frac{n}{2}(a_1 + a_n)$

d. $S_n = a_1(1-r^n)$

5 Find the sum of the first 6 terms of the series $4 + 12 + 36 + \ldots$



a. 1456

b. 1458

c. 486

d. 2916

6 Evaluate $\sum_{k=1}^{5} 2(3)^{k-1}$.

a. 242

b. 162

c. 243

d. 486

Section 3: Infinite Geometric Series

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Under what condition does an infinite geometric series $\sum_{n=1}^{\infty} a_1 r^{n-1}$ converge?

a. $|r| < 1$

b. $r > 1$

c. $|r| \leq 1$

d. $r > 0$

8 Find the sum of the infinite series $12 + 4 + \frac{4}{3} + \frac{4}{9} + \ldots$

a. 18

b. 16

c. 24

d. 15

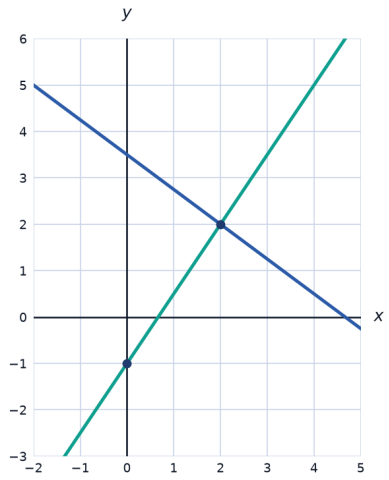
9 Find the fractional equivalent of the repeating decimal $0.\overline{45} = 0.454545\ldots$ using an infinite geometric series.

a. 5/11

b. 45/100

c. 9/22

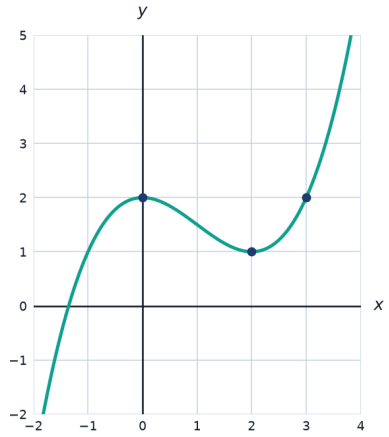
d. 45/90



Section 4: Geometric Word Problems and Annuities

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 A bouncing ball is dropped from a height of 16 ft. On each bounce, it rebounds to $3/4$ of its previous height. Find the total vertical distance the ball travels before coming to rest.



a. 112 ft

b. 64 ft

c. 128 ft

d. 96 ft

11 A company offers a starting salary of 50000 dollars with a guaranteed 4% annual raise. What is the salary in the 5th year?

a. $50000(1.04)^4 \approx 58492.93$ dollars

b. 60000 dollars

c. 58000 dollars

d. $50000(1.04)^5$

12 An ordinary annuity pays 100 dollars at the end of each year for 10 years at 5% annual interest. Which expression represents its future value?

a. $100\left[\frac{(1.05)^{10}-1}{0.05}\right]$

b. $100(1.05)^{10}$

c. $1000(1.05)$

d. $\frac{100}{0.05}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647